

Random Variable Cheat Sheet v2.0

Advanced Statistics

$$E[X] = \mu_X = \sum_{i=1}^n p(x_i) \cdot x_i$$

$$E[f(x)] = \sum_{i=1}^n p(x_i) \cdot f(x_i)$$

$$Var[X] = E[(X - \mu_X)^2] = E[X^2] - (E[X])^2 = \sigma_x^2 = \sum_{i=1}^n p(x_i) \cdot (x_i - \mu_X)^2$$

Continuous Random Variables

$\rho(X)$ = “the probability density of X” (for continuous random variables)

$F(X)$ = “the cumulative distribution of X”

$$F[x] = P(X \leq x)$$

Sum of Random Variables

$$E[X + Y] = E[X] + E[Y] \text{ (expected values add)}$$

$$\text{when X and Y are independent: } Var[X + Y] = Var[X] + Var[Y]$$

$$\text{more generally: } Var[X \pm Y] = Var[X] + Var[Y] \pm 2 \cdot Cov[X, Y]$$

where:

$$Cov[X, Y] = E[X \cdot Y] - E[X] \cdot E[Y]$$

$$Corr[X, Y] = \frac{Cov[X, Y]}{\sigma_x \cdot \sigma_y}$$

Linear Transformations of Random Variables:

$$\text{If } Y = aX + b \text{ then } E[Y] = aE[X] + b \text{ and } Var[Y] = a^2Var[X]$$

Binomial Random Variables

$$X \sim \text{binomial}(n, p)$$

X is the number of successes in n independent trials each with p probability of success

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np$$

$$Var[X] = np(1 - p)$$

Uniform Random Variables

$$X \sim \text{unif}(a,b)$$

X is a randomly selected value from the interval (a,b) where all values have the same probability density.

$$\rho[x] = \frac{1}{b-a}$$

$$E[X] = \frac{a+b}{2}$$

$$\text{Var}[X] = \frac{(b-a)^2}{12}$$

Normal Random Variables

$X \sim N(\mu, \sigma^2)$, where μ is the mean and σ is the standard deviation

$$\rho(x) = \sqrt{\frac{1}{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Central Limit Theorem

The probability distribution of means or sums of a large number of independent draws from ANY distribution (with a finite expected value) can be approximated by a normal distribution. Outcomes that are affected by a large number of individually small, mostly independent, factors tend to be roughly normally distributed.

Z-Scores

The Z-score of an observation is the number of standard deviations it falls above or below the mean. We compute the Z-score for an observation x that follows a distribution with mean μ and standard deviation σ using

$$Z = \frac{x - \mu}{\sigma}$$

$$Z \sim N(0,1)$$

$$\rho(z) = \sqrt{\frac{1}{2\pi}} e^{-\frac{z^2}{2}}$$

$F(z) = \dots$ look on a standard normal table!... or use pnorm in R

Exponential Random Variables

$X \sim \text{exp}(\lambda)$ where λ is the rate at which events occur in a process in which events occur continuously and independently.

$$\rho(X) = \lambda e^{-\lambda x}$$

$$F(x) = 1 - e^{-\lambda x}$$

$$E[X] = \frac{1}{\lambda}$$

$$\text{Var}[X] = \frac{1}{\lambda^2}$$

Poisson Random Variables

$X \sim \text{Poisson}(\lambda)$ where λ is the mean number of events that occur in a process in which events occur continuously and independently at a constant average rate.

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

$$E[X] = \lambda$$

$$\text{Var}[X] = \lambda$$